

MAHSA KHODABANDEHLOU

Data Analyst | Statistical Modeling | Machine Learning

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[LinkedIn](#) | [GitHub](#) | [Medium](#)

Education

B.Sc. in Statistics

Allameh Tabataba'i University, Tehran, Iran

09/2022 – 09/2026

Technical Skills

Programming & Tools	Python (Pandas, NumPy, Scikit-learn, Statsmodels), R (dplyr, ggplot2, caret), SQL, Docker, Git/GitHub, Jupyter Notebook
Data Analysis & ML	Linear/Logistic/Multinomial Regression, Random Forest, KNN, SVM, Decision Trees, Clustering, Data Mining, Feature Engineering
Time Series & Statistics	ARIMA, VAR, Stationarity Testing, Granger Causality, ANOVA, Hypothesis Testing, Statistical Inference, Model Diagnostics
Dimensionality Reduction	PCA, Kernel PCA, t-SNE, UMAP, Autoencoders
NLP	Text Preprocessing, TF-IDF, Sentiment Analysis, Multinomial Naive Bayes
Visualization & BI	Power BI, Advanced Excel, Matplotlib, Seaborn

Selected Projects

Data Pipeline – Kafka, PostgreSQL and NocoDB

- Built a containerized data pipeline using Kafka for stream processing, PostgreSQL with WAL-based backup/archiving, and NocoDB for data management, orchestrated via Docker Compose.

Dimensionality Reduction – Comparative Study

- Compared PCA, Kernel PCA, t-SNE, UMAP, and Autoencoders using quantitative and visual evaluation criteria.
- Diagnosed and fixed a methodological issue in the Autoencoder training pipeline, improving the Trustworthiness score from 0.61 to 0.92.

Stock Market Time Series Forecasting – Apple Inc.

- Analyzed AAPL Open, Close, and Volume data (2015–2024) using differencing, ADF stationarity tests, and VAR models with systematic lag selection.
- Performed Granger causality analysis and generated 30-day forecasts with out-of-sample / walk-forward evaluation.

Data Mining – Classification & Clustering

- Built an end-to-end classification pipeline (Logistic Regression, Decision Tree, SVM, KNN) with temporal train/test splitting and feature engineering.
- Applied and compared K-Means, DBSCAN, OPTICS, and BIRCH clustering structures.

Sentiment Analysis of Product Reviews

- Built an NLP pipeline (TF-IDF, unigrams/bigrams) comparing Multinomial Naive Bayes and Logistic Regression with 5-fold cross-validation.
- Evaluated using Accuracy, Precision, Recall, and F1-score; deployed trained model and vectorizer with Joblib.

Data Storytelling Dashboard – Iran Population and Labor Force

- Built an interactive dashboard visualizing a decade of demographic and labor-market shifts in Iran using HTML, CSS, JavaScript, and Chart.js, with 15+ charts synthesized from multiple official sources.

Experience

Python Instructor

- Taught Python fundamentals and introductory data analysis, including practical use of Pandas and NumPy.

1+ year

Courses & Certifications

Data Engineering – Filoger	12/2023 – 03/2024
Elements of AI & Building AI – University of Helsinki	02/2026
Python in Excel – LinkedIn Learning	2025
One Million Prompts – Dubai Future Foundation	09/2026
Algorithm LevelUp – Quera	08/2025

Achievements

- Ranked **141st nationwide among 2,000+ participants** in Quera Seven Adventures (2023).
- Received a **Merit Scholarship** for the Data Engineering program at Filoger (2023).

Languages

Persian: Native

English: Professional Working Proficiency